Question:

Lee used to be able to run the 400-meter hurdles two seconds faster than Gerald would run the 400-meter hurdles. But Gerald changed his diet, which improved his speed by 10%. If Lee runs the 400-meter hurdles in 38 seconds, how fast can Gerald, with his improved diet, run the 400-meter hurdles, in seconds?

Reasoning + Answer:

<think>To solve this problem, let's first determine how long it took Lee to run the 400-meter hurdles before his change in diet. According to the problem, Lee can run the 400-meter hurdles 2 seconds faster than Gerald. Since we know that Lee runs in 38 seconds after his diet change, we can use this information to find out how long it took Lee before the change. We can set up a simple subtraction to find the original time it took Lee: \(38 \text{seconds} + 2 \text{seconds} = 40 \text{seconds} \). So, before the change, Lee took 40 seconds to complete the 400-meter hurdles. Next, we need to find out how long it would take Gerald to run the 400-meter hurdles after his diet change. The problem states that Gerald's speed improved by 10% after he changed his diet. To find Gerald's new time, we first need to determine Gerald's original time before the diet change. Since Lee was running 2 seconds faster than Gerald, and we've determined that Lee's original time was 40 seconds, we can say Gerald's original time was \(40 + 2 = 42 \) seconds. Now, since Gerald's speed improved by 10% after changing his diet, we need to calculate 10% of his original time to find out how much his new time would be. To do this, we calculate \(42 \times 0.10 = 4.2 \) seconds. We subtract this improvement from Gerald's original time to find his new time: \(42 - 4.2 = 37.8 \) seconds.</think><answer> 37.8</answer>